SMART MANUFACTURING AI

Visual Quality Inspection & Automated Defect Intelligence Platform

RESULT: REVIEW REQUIRED

REVIEW REQUIRED

Inspection ID:	INSP-1793369898-EFA1F3	**Timestamp:**	2026-10-30 19:48:19
Component Name:	Hydraulic component	**Similarity Score (SSIM):**	88.69%
Total Defect Regions:	5	**Affected Surface Area:**	0.099%
High-Severity Defects:	0	**Processing Duration:**	297.4 ms

Visual Evidence & Comparative Computer Vision Analysis

Figure A: Reference (Good Component)

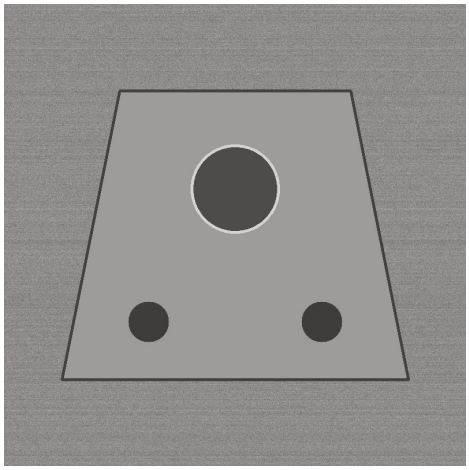


Figure B: Inspected Component

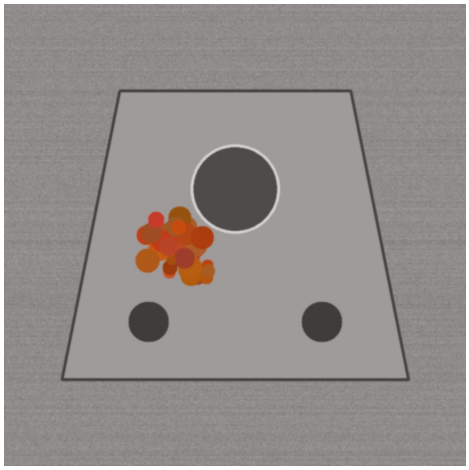


Figure C: SSIM Difference Heatmap

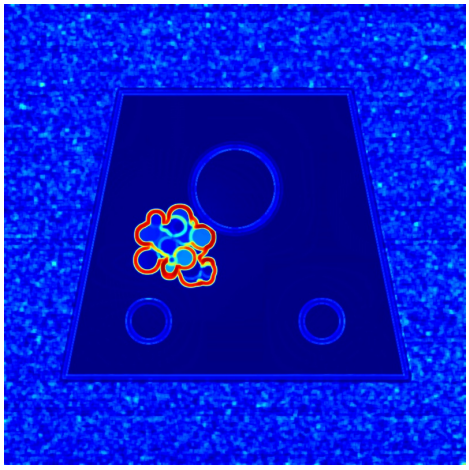
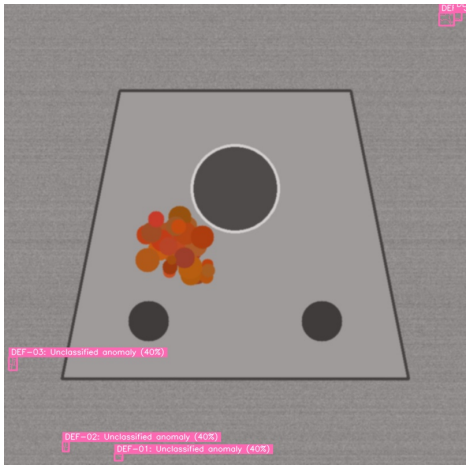


Figure D: Defect Highlighted & Annotated



Detected Defects & Anomaly Registry

Defect ID	Category	Severity	Score	Location (X, Y, W, H)	Area (% / px)	Description

DEF-01	**Unclassified anomaly**	**LOW**	0.40	(191, 780) 14x12	0.02% (101px)	Unclassified photometric anomaly requiring manual inspector verification.
DEF-02	**Unclassified anomaly**	**LOW**	0.40	(101, 759) 11x17	0.01% (80px)	Unclassified photometric anomaly requiring manual inspector verification.
DEF-03	**Unclassified anomaly**	**LOW**	0.40	(8, 612) 14x23	0.01% (88px)	Unclassified photometric anomaly requiring manual inspector verification.
DEF-04	**Unclassified anomaly**	**LOW**	0.40	(753, 15) 26x22	0.03% (207px)	Unclassified photometric anomaly requiring manual inspector verification.
DEF-05	**Unclassified anomaly**	**LOW**	0.40	(779, 8) 13x20	0.02% (154px)	Unclassified photometric anomaly requiring manual inspector verification.

AUTOMATED QUALITY ENGINEERING RECOMMENDATIONS:

• Manual inspection recommended: Examine highlighted anomaly area under uniform 1000-lux inspection lighting.

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